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Stationarity of GARCH processes and of some nonnegative time series

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Two time series models are considered: GARCH processes and generalized multivariate autoregressive equations, $X_{n+1} = A_{n+1}X_n + B_{n+1}$, with nonnegative i.i.d. coefficients. In each case, a necessary and sufficient condition ensuring the existence of a strictly stationary solution is given.

1. Introduction

In this paper, we give a necessary and sufficient condition for existence of a strictly stationary solution of two related time series equations. Namely, we consider GARCH models and generalized autoregressive models with nonnegative coefficients.

The ARCH (Autoregressive Conditional Heteroscedasticity) model was introduced in Engle (1982) to allow conditional variance of a time series to change over time as a function of the past. These series have a tendency for large changes to be followed by large changes but of unpredictable sign. This model was extended to Generalized ARCH (GARCH) time series by Bollerslev (1986). We fix a probability measure λ on $\mathbb{R}$ with mean 0 and variance 1. We say that a random variable Y is of type λ if either $Y = 0$, a.s., or if for some $\sigma > 0$ the law of $\sigma^{-1}Y$ is λ. In that case, σ^2 is the variance of Y.

Definition 1.1. A sequence of real random variables $\{Y_n,\ z \in \mathbb{Z}\}$ is said to be a GARCH(p, q) process if:

(i) *the distribution of Y_n conditional on the σ-field $\mathcal{F}_{n-1} = \sigma(Y_k,\ k \leq n-1)$ is of type λ with variance σ_n^2,*

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(ii) *there exist nonnegative constants* δ, β_i, $1 \le i \le p$, *and* α_j, $1 \le j \le q$, *such that*

$$\sigma_n^2 = \delta + \sum_{i=1}^{p} \beta_i \sigma_{n-i}^2 + \sum_{j=1}^{q} \alpha_j Y_{n-j}^2, \qquad n \in \mathbb{Z}. \tag{1}$$

The second-order properties of these models are well known. In particular Bollerslev (1986) has shown the following theorem. [We recall that a process $\{X_n\}$ is second-order, or weakly, stationary, if each X_n is square integrable and if for all $n, m \in \mathbb{Z}$, $\mathbb{E}(X_n)$ and $\operatorname{cov}(X_n, X_{n+m})$ are independent of n.]

Theorem 1.2 (Bollerslev). When $\delta > 0$, *the GARCH eq. (1) defines a second-order stationary solution if and only if*

$$\sum_{i=1}^{p} \beta_i + \sum_{j=1}^{q} \alpha_j < 1.$$

(When $\delta = 0$, $Y_n = 0$ is always a stationary solution.) But it has been empirically shown that some financial series (for instance exchange rates and interest rates) typically exhibit parameters which are not in the second-order stationarity region [see Engle and Bollerslev (1986) and references therein]. Although these series are not square-integrable, they are strictly stationary. [A process $\{X_n\}$ is strictly stationary if for all $n, m \in \mathbb{Z}$, the law of $(X_n, X_{n+1}, \ldots, X_{n+m})$ is independent of n.] Therefore the question arises to develop conditions for strict stationary of these GARCH processes. Nelson (1988) solved this problem for GARCH(1, 1) when λ is the normal law. He has also indicated that the theory of products of random matrices should be the right tool to handle the general case. Following his suggestion, we will show that the necessary and sufficient condition for stationarity is the negativity of an associated Lyapunov exponent, which is a central concept of this theory.

Let us recall the definition of the top Lyapunov exponent. Choose any norm $\|\cdot\|$ on $\mathbb{R}^d$, and define an operator norm on the set $\mathbb{M}(d)$ of $d \times d$ matrices by

$$\|M\| = \sup\{\|Mx\|/\|x\|;\ x \in \mathbb{R}^d,\ x \neq 0\},$$

for any M in $\mathbb{M}(d)$. Then, the top Lyapunov exponent associated to a sequence $\{A_n,\ n \in \mathbb{Z}\}$ of independent, identically distributed, random matri-

ces, is defined by

$$\gamma = \inf\left\{\mathbb{E}\left(\frac{1}{n+1}\log\|A_0A_{-1}\dots A_{-n}\|\right),\ n\in\mathbb{N}\right\},$$

when $\mathbb{E}(\log^+\|A_0\|)$ is finite [where $\log^+ x = \max(\log x, 0)$]. For instance, $\gamma \le \mathbb{E}(\log\|A_0\|)$, with equality when $d = 1$. It follows from the subadditive ergodic theorem [see Kingman (1973, theorem 6)] that, almost surely,

$$\gamma = \lim_{n\to+\infty}\frac{1}{n}\log\|A_0A_{-1}\dots A_{-n}\|. \tag{2}$$

Since all the norms are equivalent on $\mathbb{R}^d$, γ is independent of the norm. In general, this exponent can be difficult to compute but it is easily estimated by computer simulations, using (2). An overview of the theory of products of random matrices is presented in Cohen, Kesten, and Newman (1986).

We need to introduce the following notations. For notational convenience we can (and will) always suppose that $p, q \ge 2$, by adding some α_i or β_i equal to 0 if needed. For any $n \in \mathbb{Z}$, let $\eta_n = \sigma_n^{-1}Y_n$, and

$$\tau_n = \left(\beta_1 + \alpha_1\eta_n^2, \beta_2, \dots, \beta_{p-1}\right) \in \mathbb{R}^{p-1},$$

$$\xi_n = \left(\eta_n^2, 0, \dots, 0\right) \in \mathbb{R}^{p-1},$$

$$\alpha = (\alpha_2, \dots, \alpha_{q-1}) \in \mathbb{R}^{q-2}.$$

Define the $(p+q-1)\times(p+q-1)$ matrix A_n, written in block form, by

$$A_n = \begin{bmatrix} \tau_n & \beta_p & \alpha & \alpha_q \\ I_{p-1} & 0 & 0 & 0 \\ \xi_n & 0 & 0 & 0 \\ 0 & 0 & I_{q-2} & 0 \end{bmatrix}, \tag{3}$$

where I_{p-1} and I_{q-2} are the identity matrices of size $p-1$ and $q-2$, respectively. We will see that the random variables $\{\eta_n,\ n\in\mathbb{Z}\}$ are independent, with law λ. Therefore, the random matrices $\{A_n,\ n\in\mathbb{Z}\}$ are i.i.d. Since λ has a finite variance, all the coefficients of these matrices are integrable. This implies that $\mathbb{E}(\log^+\|A_0\|)$ is finite. Thus, the top Lyapunov exponent γ of the sequence $\{A_n,\ n\in\mathbb{Z}\}$ is well defined.

Theorem 1.3. When $\delta > 0$, the GARCH eq. (1) has a strictly stationary solution if and only if the top Lyapunov exponent γ associated with the matrices

$\{A_n, n \in \mathbb{Z}\}$ is strictly negative. Moreover, this stationary solution is ergodic. It is the only strictly stationary solution when the η_n's are given.

This theorem reduces to the result of Nelson (1988) when $p = q = 1$, since in that case $\gamma = \mathbb{E}(\log(\beta_1 + \alpha_1 \eta_n^2))$. To prove Theorem 1.3, we use the connection between GARCH processes and the following multivariate model. Let $\mathbb{M}^+(d)$ [resp. $(\mathbb{R}^+)^d$] be the set of $d \times d$ matrices (resp. d-dimensional vectors) with nonnegative coefficients.

Definition 1.4. A generalized autoregressive equation with nonnegative i.i.d. coefficients is

$$X_{n+1} = A_{n+1} X_n + B_{n+1}, \qquad n \in \mathbb{Z},$$

where $\{(A_n, B_n), n \in \mathbb{Z}\}$ is a given sequence of independent, identically distributed, random variables with values in $\mathbb{M}^+(d) \times (\mathbb{R}^+)^d$ and X_n is in $\mathbb{R}^d$.

This model is interesting in itself. It is studied, for instance, in Vervaat (1979), Gaver and Lewis (1980), Lawrance and Lewis (1980), and Hutton (1990). Under a weak condition, we show that a necessary and sufficient condition for existence of a strictly stationary, nonnegative solution of this equation is the strict negativity of the top Lyapunov exponent associated with the sequence $\{A_n, n \in \mathbb{Z}\}$.

The plan of the paper is as follows. We prove Theorem 1.3 in section 2 and study the general autoregressive model with nonnegative i.i.d. coefficients in section 3. Actually, Theorem 1.3 can be deduced from the results of section 3. But we prefer to give first a simple self-contained proof of this theorem. If we drop the condition of nonnegativity, the situation is more complicated. It has been studied in Bougerol and Picard (1990).

In the two sections, we use the following notations. When $x, y \in \mathbb{R}^d$,

$$x \leq y \quad \text{if} \quad y - x \in (\mathbb{R}^+)^d.$$

When $A, B \in \mathbb{M}(d)$,

$$A \leq B \quad \text{if} \quad B - A \in \mathbb{M}^+(d). \tag{4}$$

2. Stationarity of GARCH processes

The aim of this section is to prove Theorem 1.3 on GARCH processes. We will need the following general lemma.

Lemma 2.1. *Let* $\{A_n, n \in \mathbb{Z}\}$ *be a sequence of independent, identically distributed, random matrices such that* $\mathbb{E}(\log^+ \|A_0\|)$ *is finite. If, almost surely,*

$$\lim_{n \to +\infty} \|A_0 A_{-1} \dots A_{-n}\| = 0,$$

then the top Lyapunov exponent associated with this sequence is strictly negative.

Proof. This results immediately from the application to the transpose of A_n of Bougerol (1987, lemma 5.2) when A_n is invertible, or of its (easy) extension to the noninvertible case in Bougerol and Picard (1990).

We consider a GARCH(p,q) process as defined in Definition 1.1. Let $\eta_n = \sigma_n^{-1} Y_n$. The distribution of η_n conditional on $\mathcal{F}_{n-1}$ is λ. Therefore η_n is independent of $\mathcal{F}_{n-1}$. Since η_n is $\mathcal{F}_n$-measurable, this implies that the random variables $\{\eta_n, n \in \mathbb{Z}\}$ are independent, with law λ.

Let

$$B = {}^t(\delta, 0, \dots, 0) \in \mathbb{R}^{p+q-1}$$

(the 't' is for transpose) and

$$X_n = {}^t\left(\sigma_{n+1}^2, \dots, \sigma_{n-p+2}^2, Y_n^2, \dots, Y_{n-q+2}^2\right). \tag{5}$$

Then Y_n is a solution of (1) if and only if X_n is a solution of

$$X_{n+1} = A_{n+1} X_n + B, \qquad n \in \mathbb{Z}. \tag{6}$$

[The matrices A_n are defined by (3).]

Proof of Theorem 1.3. We suppose that there exists a strictly stationary solution $\{Y_n, n \in \mathbb{Z}\}$ of (1). Consider the process $\{X_n, n \in \mathbb{Z}\}$ defined by (5). We have for $n > 0$, using (6),

$$\begin{aligned} X_0 &= A_0 X_{-1} + B \\ &= A_0 A_{-1} X_{-2} + B + A_0 B \\ &= A_0 A_{-1} \dots A_{-n} X_{-n-1} + B + \sum_{k=0}^{n-1} A_0 \dots A_{-k} B. \end{aligned}$$

All the coefficients of A_n, X_n, and B are nonnegative. Thus, for any $n > 0$,

$$\sum_{k=0}^{n-1} A_0 \dots A_{-k} B \le X_0.$$

This shows that the series $\sum_{k=0}^{n-1} A_0 \dots A_{-k} B$ converges a.s. Therefore, $A_0 \dots A_{-n} B$ converges to zero, a.s., when $n \to +\infty$. Let (e_i) be the canonical basis of $\mathbb{R}^{p+q-1}$. We will prove that, for $1 \le i \le p+q-1$,

$$\lim_{n \to +\infty} A_0 \dots A_{-n} e_i = 0. \tag{7}$$

Since $B = \delta e_1$ and δ is different from zero, (7) holds for $i = 1$. Thus, since $A_{-n} e_p = \beta_p e_1$, we have, a.s.,

$$\lim_{n \to +\infty} A_0 \dots A_{-n+1} A_{-n} e_p = \beta_p \lim_{n \to +\infty} A_0 \dots A_{-n+1} e_1 = 0.$$

Now, if (7) holds for some j, $2 < j \le p$, then, using

$$A_{-n} e_{j-1} = \beta_{j-1} e_1 + e_j,$$

we have

$$\lim_{n \to +\infty} A_0 \dots A_{-n+1} A_{-n} e_{j-1} = \lim_{n \to +\infty} A_0 \dots A_{-n+1} (\beta_{j-1} e_1 + e_j) = 0.$$

Therefore, by a backward recursion, (7) holds for any $j \le p$. Using the same argument for $e_{p+q-1}, \dots, e_{p+1}$ and the relations

$$A_{-n} e_{p+q-1} = \alpha_q e_1, \qquad A_{-n} e_{p+j-1} = \alpha_j e_1 + e_{p+j},$$

$(2 \le j \le q-1)$, we see that (7) holds for all the e_i's. Thus

$$\lim_{n \to +\infty} A_0 \dots A_{-n} = 0, \quad \text{a.s.}$$

We now use Lemma 2.1, to conclude that the top Lyapunov exponent associated with the matrices A_n is strictly negative. This proves the 'only if' part of the theorem.

Let us show the 'if' part. We suppose now that the exponent γ is strictly negative. Then (2) implies that the series

$$\sum_{k=0}^{\infty} A_n A_{n-1} \dots A_{n-k} B$$

converges a.s. for any n. We define a sequence $\{X_n, n \in \mathbb{Z}\}$ by

$$X_n = B + \sum_{k=0}^{\infty} A_n A_{n-1} \dots A_{n-k} B.$$

This sequence is a nonnegative solution of (6). Let $\sigma_n = \sqrt{X_{n-1}^1}$, where X_{n-1}^1 is the first component of the vector X_{n-1}. Then $Y_n = \sigma_n \eta_n$ is a solution of the GARCH model (1). The process $\{(A_n, \eta_n),\ n \in \mathbb{Z}\}$ is strictly stationary and ergodic, and we can write $Y_n = F(\eta_n, A_n, A_{n-1}, A_{n-2}, \ldots)$ for some measurable function F independent of n. Therefore $\{Y_n,\ n \in \mathbb{Z}\}$ is a strictly stationary and ergodic process, solution of (1).

Finally, let (Z_n) be another strictly stationary solution of (6). The relation, for $n \geq 0$,

$$\begin{aligned} \|X_0 - Z_0\| &= \|A_0 \ldots A_{-n}(X_{-n-1} - Z_{-n-1})\| \\ &\leq \|A_0 \ldots A_{-n}\| \, \|X_{-n-1} - Z_{-n-1}\|, \end{aligned}$$

the fact that $\|A_0 \ldots A_{-n}\|$ converges to 0, a.s., and the fact that the law of $(X_{-n-1} - Z_{-n-1})$ is independent of n, imply that $X_0 - Z_0$ converges to 0 in probability. Therefore, $X_0 = Z_0$. This shows that (1) has a unique solution, once the η_n's are given, and ends the proof of the theorem.

As an application, let us prove the following corollary on the so-called integrated GARCH. For convenience, we use stronger hypotheses than needed.

Corollary 2.2. We suppose that the support of λ is not bounded, that $\lambda(\{0\}) = 0$, and that all the coefficients α_i and β_i are positive. Then, if $\sum_{i=1}^{p} \beta_i + \sum_{j=1}^{q} \alpha_j = 1$, the GARCH eq. (1) has a unique strictly stationary solution.

Proof. Arguing by recursion on q and expanding the determinant with respect to the last column, it is not difficult to check that

$$\operatorname{Det}(z\,\mathrm{Id} - \mathbb{E}(A_1)) = z^{p+q-1}\left(1 - \sum_{j=1}^{q} \alpha_j z^{-j} - \sum_{i=1}^{p} \beta_i z^{-i}\right).$$

Now it follows from the inequality $|a - b| \geq |(|a| - |b|)|$ that, if $|z| > 1$, then

$$\left|\operatorname{Det}(z\,\mathrm{Id} - \mathbb{E}(A_1))\right| > 1 - \sum_{j=1}^{q} \alpha_j - \sum_{i=1}^{p} \beta_i. \tag{8}$$

Since the right-hand side is zero and since $\operatorname{Det}(\mathrm{Id} - \mathbb{E}(A_1)) = 0$, (8) implies that the spectral radius ρ of the matrix $\mathbb{E}(A_1)$ is 1. On the other hand, almost surely, all the coefficients of the matrix $A_2 A_1$ are positive and A_1 has no zero column nor zero row. Since A_1 is not a.s. bounded, these properties imply by Kesten and Spitzer (1984, theorem 2) that the top Lyapunov exponent γ satisfies $\gamma < \log \rho$. So that $\gamma < 0$ and the corollary follows from Theorem 1.3.

Remark. When $\sum_{i=1}^{p}\beta_i + \sum_{j=1}^{q}\alpha_j < 1$, it follows from Bollerslev (1986) that (1) has a strictly stationary solution. This is also a consequence of Theorem 1.3: indeed, (8) implies that the spectral radius ρ of $\mathbb{E}(A_1)$ satisfies $\rho < 1$. Since, it is always true that $\gamma \leq \log \rho$ [see Kesten and Spitzer (1984, (1.4))], we see that $\gamma < 0$.

Corollary 2.3. When $\delta > 0$, if the GARCH model (1) has a strictly stationary solution, then $\sum_{i=1}^{p}\beta_i < 1$.

Proof. Let A be the matrix which is obtained by replacing η_n by 0 in the matrices A_n defined by (3). Since for each k in $\mathbb{Z}$, $A_k \geq A$, we have

$$A_0 \ldots A_{-n} \geq A^{n+1}.$$

Therefore, the top Lyapunov exponent is greater than $\log \rho$. By Theorem 1.3, this exponent is strictly negative, thus $\rho < 1$. On the other hand, it is easily seen that the characteristic polynomial of A is

$$\mathrm{Det}(z\,\mathrm{Id} - A) = \lambda^{p+q-1}\left(1 + \sum_{i=1}^{p}\beta_i z^{-i}\right).$$

We see that the continuous function $f(x) = (1 - \sum_{i=1}^{p}\beta_i x^i)$ has no zero when $0 \leq x \leq 1$. Since $f(0) = 1$, this implies that $f(1) = 1 - \sum_{i=1}^{p}\beta_i > 0$.

3. Autoregressive model with nonnegative i.i.d. coefficients

We now study the existence of nonnegative strictly stationary solutions $\{X_n,\ n \in \mathbb{Z}\}$ of the autoregressive equation

$$X_{n+1} = A_{n+1}X_n + B_{n+1}, \qquad n \in \mathbb{Z}, \tag{9}$$

where $\{(A_n, B_n),\ n \in \mathbb{Z}\}$ is a sequence of i.i.d. random variables in $\mathbb{M}^+(d) \times (\mathbb{R}^+)^d$. In order to avoid some degenerate cases, we introduce the following condition [where (e_i) denotes the canonical basis of $\mathbb{R}^d$].

Definition 3.1. The autoregressive eq. (9) satisfies Condition (C), if there does not exist a strict subset I of $\{1, \ldots, d\}$ such that, if V is the linear span of $\{e_i,\ i \in I\}$, then

$$A_n V + B_n \subset V, \quad a.s.$$

(By strict we mean that I is different from $\{1, \ldots, d\}$. It can be empty, and in that case V reduces to $\{0\}$.) In other words, Condition (C) holds if there does not exist any permutation σ of $\{1, \ldots, d\}$ such that in the basis

$(e_{\sigma(1)},\dots,e_{\sigma(d)})$, A_n and B_n can be simultaneously written in a block form as

$$A_n = \begin{bmatrix} E_n & 0 \\ F_n & G_n \end{bmatrix}, \qquad B_n = \begin{bmatrix} 0 \\ H_n \end{bmatrix}.$$

For instance (C) holds if all the coefficients of A_n (or B_n) are strictly positive with nonzero probability.

Theorem 3.2. We suppose that Condition (C) holds and that $\mathbb{E}(\log^+\|A_n\|)$ *is finite.*

A. If eq. (9) has a strictly stationary, nonnegative, solution, then the top Lyapunov exponent γ *associated with the matrices* $\{A_n,\ n\in\mathbb{Z}\}$ *is strictly negative.*

B. Conversely, if $\mathbb{E}(\log^+\|B_0\|)$ *is finite and if* $\gamma<0$, *then for all* $n\in\mathbb{Z}$, *the series*

$$X_n = B_n + \sum_{k=1}^{+\infty} A_n A_{n-1}\dots A_{n-k+1}B_{n-k}$$

converges a.s. and the process $\{X_n,\ n\in\mathbb{Z}\}$ *is the unique strictly stationary solution of (9).*

Proof. We first prove the first statement. The beginning of this proof is similar to that of Theorem 1.3, but we repeat the argument for the convenience of the reader. Let $\{X_n,\ n\in\mathbb{Z}\}$ be a strictly stationary nonnegative solution of (9). For any positive integer n, write

$$A^{(n)} = A_0 A_{-1}\dots A_{-n}$$

and

$$B^{(n)} = B_0 + \sum_{k=1}^{n} A_0\dots A_{-k+1}B_{-k}.$$

Using repeatedly eq. (9), we have

$$X_0 = A^{(n)}X_{-n-1} + B^{(n)}.$$

All the coefficients of A_n, X_n, and B_n are nonnegative. Therefore $B^{(n)}\le X_0$. This shows that the sequence $\{B^{(n)},\ n\in\mathbb{N}\}$ is a bounded series with nonnegative coefficients. Hence, it converges, a.s., to a random variable

$$Z = B_0 + \sum_{k=1}^{+\infty} A_0\dots A_{-k+1}B_{-k}.$$

Let m be the law of Z and μ be the law of (A_n, B_n). We have

$$A_1 Z + B_1 = B_1 + \sum_{k=1}^{+\infty} A_1 A_0 \dots A_{-k+2} B_{-k+1}.$$

Since (A_n, B_n) is stationary, the law of $A_1 Z + B_1$ is m. Therefore, using also the independence of (A_1, B_1) and Z, we may write, for any bounded Borel function $f: \mathbb{R}^d \to \mathbb{R}$,

$$\begin{aligned} \int f(x)\, \mathrm{d}m(x) &= \mathbb{E}(f(A_1 Z + B_1)) \\ &= \mathbb{E}\left(\int f(A_1 x + B_1)\, \mathrm{d}m(x) \right) \\ &= \iint f(Ax + B)\, \mathrm{d}m(x)\, \mathrm{d}\mu(A, B). \end{aligned} \tag{10}$$

For $n \in \mathbb{N}$, let

$$Y_n = \int f(A^{(n)} x + B^{(n)})\, \mathrm{d}m(x),$$

and let $\mathcal{F}_n$ be the σ-field

$$\mathcal{F}_n = \sigma(A^{(k)}, B^{(k)}, 0 \le k \le n).$$

We have, using (10),

$$\begin{aligned} \mathbb{E}(Y_{n+1} | \mathcal{F}_n) &= \mathbb{E}\left(\int f(A^{(n)}(A_{-n-1} x + B_{-n-1}) + B^{(n)})\, \mathrm{d}m(x) | \mathcal{F}_n \right) \\ &= \iint f(A^{(n)}(Ax + B) + B^{(n)})\, \mathrm{d}m(x)\, \mathrm{d}\mu(A, B) \\ &= \int f(A^{(n)} x + B^{(n)})\, \mathrm{d}m(x) \\ &= Y_n. \end{aligned}$$

Moreover, Y_n is uniformly bounded, since f is bounded. Thus $\{Y_n, n \in \mathbb{N}\}$ is a bounded martingale. Therefore it converges, a.s. and in L^1, to some random variable W, and

$$\mathbb{E}(W) = \mathbb{E}(Y_0) = \int f(x)\, \mathrm{d}m(x) = \mathbb{E}(f(Z)). \tag{11}$$

Now consider d continuous, increasing, bounded functions $f_1, \dots, f_d$: $\mathbb{R} \to \mathbb{R}^+$ and the function f: $\mathbb{R}^d \to \mathbb{R}^+$, defined by

$$f(x_1, \dots, x_d) = f_1(x_1) \dots f_d(x_d).$$

Then,

$$f(B^{(n)}) \leq \int f(A^{(n)}x + B^{(n)})\, \mathrm{d}m(x) = Y_n.$$

Since $B^{(n)}$ converges, a.s., to Z and Y_n converges, a.s., to W, we obtain

$$f(Z) \leq W.$$

This inequality together with (11) proves that, a.s., $f(Z) = W$ and therefore

$$f(Z) = \lim_{n \to +\infty} \int f(A^{(n)}x + B^{(n)})\, \mathrm{d}m(x). \tag{12}$$

We will show that $A^{(n)}$ converges, a.s., to 0 when $n \to +\infty$. We consider $A^{(n)}$ as an element of $[0, +\infty]^{d^2}$ and denote by $A_{ij}^{(n)}$ the coefficients of this matrix. From any sequence of integers, one can extract a subsequence $\{n_k,\ k \in \mathbb{N}\}$ such that, for $1 \leq i, j \leq d$, either $A_{ij}^{(n_k)}$ converges to a real number, or $A_{ij}^{(n_k)}$ increases to $+\infty$, when $k \to +\infty$. In each case, we denote the limit by $A_{ij}^{(\infty)}$. In view of the definition of f, this function has an obvious extension to $[0, +\infty]^d$ and

$$\lim_{k \to +\infty} f(A^{(n_k)}x + B^{(n_k)}) = f(A^{(\infty)}x + Z).$$

This implies, by (12), that

$$f(Z) = \int f(A^{(\infty)}x + Z)\, \mathrm{d}m(x).$$

But, since f is nondecreasing, $f(Z) \leq f(A^{(\infty)}x + Z)$. Thus, for almost all x with respect to m, $f(A^{(\infty)}x + Z)) = f(Z)$. Since this is true for all functions f of the given form, we see that, for almost every x,

$$A^{(\infty)}x = 0, \quad \text{a.s.} \tag{13}$$

Let us show that $A^{(\infty)} = 0$. Arguing by contradiction, we suppose that there exists a coefficient $A_{kj}^{(\infty)}$ of $A^{(\infty)}$ which is not a.s. zero. Since $(A^{(\infty)}x)_k = \sum_{i=1}^d A_{ki}^{(\infty)} x_i$ and since all the coefficients are nonnegative, (13)

implies that

$$m\{x \in \mathbb{R}^d : x_j = 0\} = 1. \tag{14}$$

For any subset I of $\{1, \dots, d\}$, let

$$V_I = \{x \in (\mathbb{R}^+)^d ; x_i = 0, \forall i \in I\},$$

and let

$$\vartheta = \{V_I, I \subset \{1, \dots, d\}\}.$$

A convex subset V of $(\mathbb{R}^+)^d$ is in ϑ if and only if, for any diagonal matrix Λ in $\mathbb{M}^+(d)$, ΛV is contained in V [indeed, if a convex subset V of $(\mathbb{R}^+)^d$ satisfies this condition, then $V = V_I$, with $I = \{i \in \{1, \dots, d\} : x_i = 0, \forall x \in V\}$]. Let I be a maximal subset of $\{1, \dots, d\}$ such that $m(V_I) = 1$ and

$$V = \{x \in (\mathbb{R}^+)^d, A_1 x + B_1 \in V_I, \text{a.s.}\}.$$

Let us show that V is in ϑ. If $x \in V$ and $i \in I$, we have $\sum_{j=1}^d (A_1)_{ij} x_j + (B_1)_i = 0$, a.s. As all the coefficients are nonnegative, necessarily for any j, $(A_1)_{ij} x_j = 0$ and $(B_1)_i = 0$. Thus for any diagonal matrix Λ in $\mathbb{M}^+(d)$, $A_1 \Lambda x + B_1 \in V_I$, and therefore $\Lambda x \in V$. The convexity of V is verified in the same way. By construction, V_I is the smallest element of ϑ such that $m(V_I) = 1$. Let $\mathbb{1}_{V_I}$ be the indicator function of the set V_I. Then

$$1 = m(V_I) = \int \mathbb{1}_{V_I}(x)\, \mathrm{d}m(x)$$

$$= \mathbb{E} \int \mathbb{1}_{V_I}(A_1 x + B_1)\, \mathrm{d}m(x) = m(V).$$

Therefore, V_I is contained in V, which implies that $A_1 V_I + B_1$ is contained in V_I, a.s. But (14) shows that $m(V_{\{j\}}) = 1$, so that V_I is not $(\mathbb{R}^+)^d$. This is a contradiction to Condition (C). This proves that $A^{(\infty)} = 0$. Therefore $A^{(n)}$ converges to 0, which implies that the top Lyapunov exponent associated with the matrices $\{A_n, n \in \mathbb{Z}\}$ is strictly negative (by Lemma 2.1). This ends the proof of part A.

The converse is proved in Brandt (1986) in the one-dimensional case, but his proof is easily extended to the multivariate situation, using (2).

Finally let us remark that the situation is different if we drop the condition of nonnegativity on the stationary solution of the autoregressive model (9).

Indeed, consider the two following one-dimensional examples:

$$X_{n+1} = \varepsilon_n X_n + (\varepsilon_n - 1), \tag{15}$$

$$Y_{n+1} = 2Y_n + \varepsilon_n, \tag{16}$$

where $\{\varepsilon_n,\ n \in \mathbb{Z}\}$ is an arbitrary sequence of i.i.d. bounded random variables greater than 1. Then $X_n = -1$ and $Y_n = -\sum_{k=0}^{+\infty} 2^{-(k+1)}\varepsilon_{n+k}$ are stationary solutions of (15) and (16), respectively, although the associated exponents are positive. This general situation is studied in Bougerol and Picard (1990).

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